AGRICULTURE AND DEVELOPMENT

The learner gives written or oral advice to a certain community or country on appropriate farming types and techniques to be adopted to promote development through appraising the different types and methods of farming.

a) Evaluate plans and strategies aimed at promoting agriculture in Uganda and other developing countries.

- Understand the meaning and broad types of agriculture: arable and livestock farming.
- Contribution of agriculture to the development of Uganda, presenting both the advantages and disadvantages.
- Comparing the importance of agriculture in their economies: one from a developing country and another from a developed country.
- Debate whether African countries should invest more or less in agriculture to achieve faster economic development and improved human welfare.

AGRICULTURE

Agriculture is the science of growing crops and rearing of livestock.

Agriculture is the major activity in many countries contributing a large percentage of the export earnings.

Agricultures is largely divided into two:

Arable Farming

This is the production of crops for food or other purposes. It requires fertile soil, balanced moisture (not too dry or wet), warm climate, and suitable machinery. Arable farming shrives best on flat or gently sloping land.

Examples: Growing wheat, maize, rice, vegetables, and fruits.

Livestock Farming

This is the raising animals for products like meat, milk, eggs, wool, or hides.

It thrives in areas with steep slopes, poor soil, or cold climates where crops struggle. The suggest land is encouraged to have pastures, grasslands, or other areas suitable for grazing.

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Examples: Raising cattle, sheep, goats, pigs, poultry, or other animals.

Question 1

Citing examples from your local communities, explain the importance of the above factors in running of both types of agriculture.

Question 2

As a geography student write an essay to be printed in New Vision newspapers highlighting the things Uganda has managed to archive because of its agricultural production also the losses it has suffered because of the same venture. (your essay should be backed up by statistics and examples from the local communities).

A COMPARATIVE STUDY OF THE IMPORTANCE OF AGRICULTURE IN DEVELOPMENT

Agriculture serves as a powerful diagnostic tool for measuring development because it reflects a society's economic structure, technological advancement, social equity, and environmental sustainability. Its transformation is deeply intertwined with broader development stages. By tracking agriculture's evolution—from labor-intensive subsistence to capital-intensive, market-driven systems—policymakers can gauge progress, identify bottlenecks (e.g., lack of credit, weak infrastructure), and prioritize interventions for inclusive growth.

Here's how agriculture functions as a development measurement tool:

- Tracking economic structural transformation in terms of share of GDP. A declining share of agriculture in GDP (e.g., from 50% to <10%) typically signals economic diversification and industrialization (e.g., South Korea, Brazil). A drop in agricultural employment (e.g., from 80% to 20%) indicates movement toward manufacturing/services, higher productivity, and urbanization. Rising yields (tons/hectare) and value-added per worker reflect technological adoption (improved seeds, irrigation, mechanization) and efficiency.
- Assessing food security and nutrition: in terms of caloric Sufficiency. Ability to produce staple crops (maize, rice, wheat) to meet national needs reduces reliance on imports. Shift from staple-only farming to fruits, vegetables, dairy, and protein sources signals improved nutrition (e.g.,

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reducing stunting). Irrigation coverage (% of cropland) and crop diversity measure capacity to withstand climate shocks.

- Evaluating poverty reduction and rural development. Rising farm incomes (from cash crops, horticulture, dairy) directly reduce rural poverty. Formal land titles (especially for women) correlate with investment in soil health and productivity. Women's access to credit, inputs, and decision-making power in farming reflects inclusive development.

- Measuring technological advancement and innovation by use of fertilizers (kg/ha), improved seed penetration, and mechanization levels indicate modernization. Use of mobile apps for market prices, weather data, or Fintech (e.g., M-Pesa for payments) reflects tech adaptation. Public/private spending on agricultural R&D (e.g., drought-resistant crops) signals long-term commitment.

- Global Market Integration in terms of export diversification: Moving from raw commodities (coffee beans) to processed goods (roasted coffee) or high-value exports (flowers, avocados) signals value-chain upgrading. Meeting international phyto-sanitary/SPS requirements demonstrates institutional capacity.

Agriculture is a mirror of development:

- Pre-transition economies rely on subsistence farming, low productivity, and high rural poverty.
- Emerging economies show rising yields, diversification, and declining farm labor.
- Developed economies feature high-tech, sustainable agriculture with minimal GDP/labor shares.

DEVELOPED COUNTRY VS A DEVELOPING COUNTRY

A CASE STUDY OF RUSSIA VS ETHIOPIA

RUSSIA'S AGRICULTURAL SECTOR

Russia is a decisive force in global food security, leveraging its unparalleled 120 million hectares of arable land **the world's largest reserve** anchored by the fertile "**chernozem"** (black earth) zones. Since 2014, targeted state subsidies, import bans, and corporate consolidation have transformed it

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from a net importer into the planet's top agricultural exporter by value in key commodities. It supplies ~25% of globally traded wheat (exporting 45–60 million metric tons annually), ~20% of barley, and ~30% of sunflower oil**, alongside massive volumes of corn, rapeseed, and fish. This export dominance – generating $40–45 billion annually pre-2022 – positions Russian grain as a critical stabilizer (or disruptor) of international markets, especially for import-dependent regions like Africa and the Middle East.

Domestically, agriculture contributes ~4–5% of Russia's GDP but underpins rural employment and 95%+ self-sufficiency in grains. The sector operates through a dual model: mega-agroholdings (like Miratorg and Rusagro) control ~75% of grain/oilseed output on vast mechanized farms, while 13 million small household plots ("dachas") supply 40% of vegetables, 80% of fruit, and 50% of milk. Despite harsh climate constraints (only 15% of land is high-yielding "chernozem"), Russia’s post-2022 role has intensified: it exported 60 million tons of grain in 2023 amid sanctions, weaponizing shipments via the Black Sea Initiative negotiations. With 1.5 billion+ people globally reliant on Russian/Ukrainian grain, its agricultural might remains central to its geopolitical strategy and global food stability.

Key statistics reinforcing Russia’s global role

1. **Grain exports**: Wheat: 25% of global exports (2023: 47.5 million tons). Barley: 18–20% of global exports. Sunflower Oil: 25–30% of global exports (pre-war: Russia + Ukraine = 75%).

2. **Land and production scale**: Arable Land: 120+ million ha (largest globally; USA: 160m ha, but only 40% as productive per hectare). Grain Harvests: 140–150 million tons annually (2023: 144m tons).

3. **Economic impact**: Export Revenue: Peaked at $41.6 billion in 2021 (UN Comtrade).

Global Dependency: 50+ countries rely on Russia/Ukraine for >30% of wheat imports (FAO).

4. **Geopolitical leverage**: Black Sea Grain Initiative (2022–2023) shipped 33 million tons of Ukrainian/Russian grain to 45+ countries. Post-sanctions, Russia redirected 70% of grain exports to Africa, Asia, and the Middle East.

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(Sources: FAO, USDA, UN Comtrade, Russian Ministry of Agriculture, World Bank)

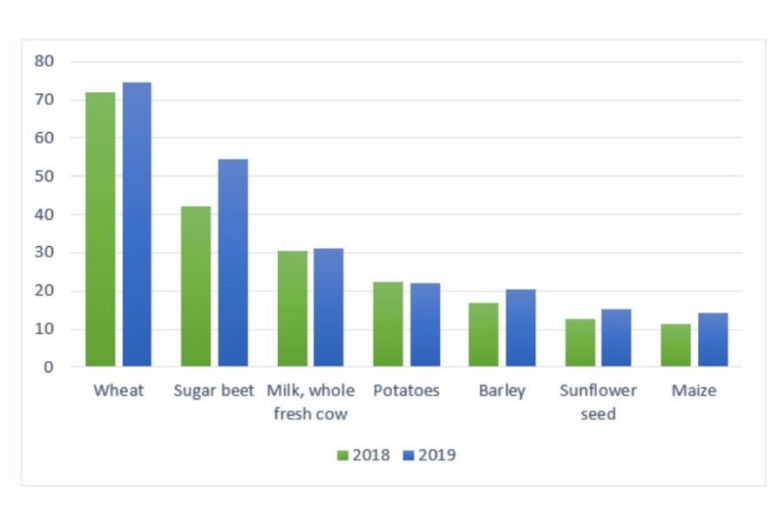


Figure 1showing production of agricultural products in Russia

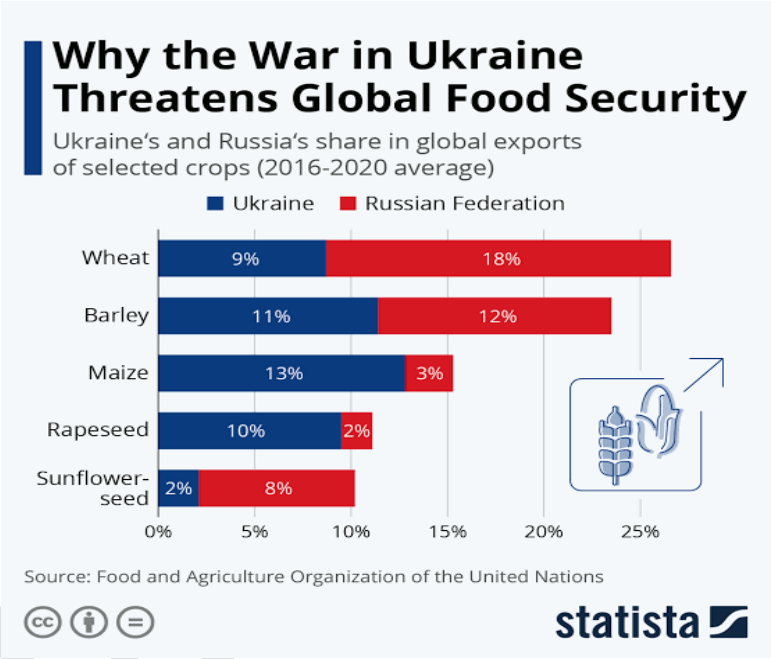


Figure 2 showing the role of Russia as a leading food source in the global food security

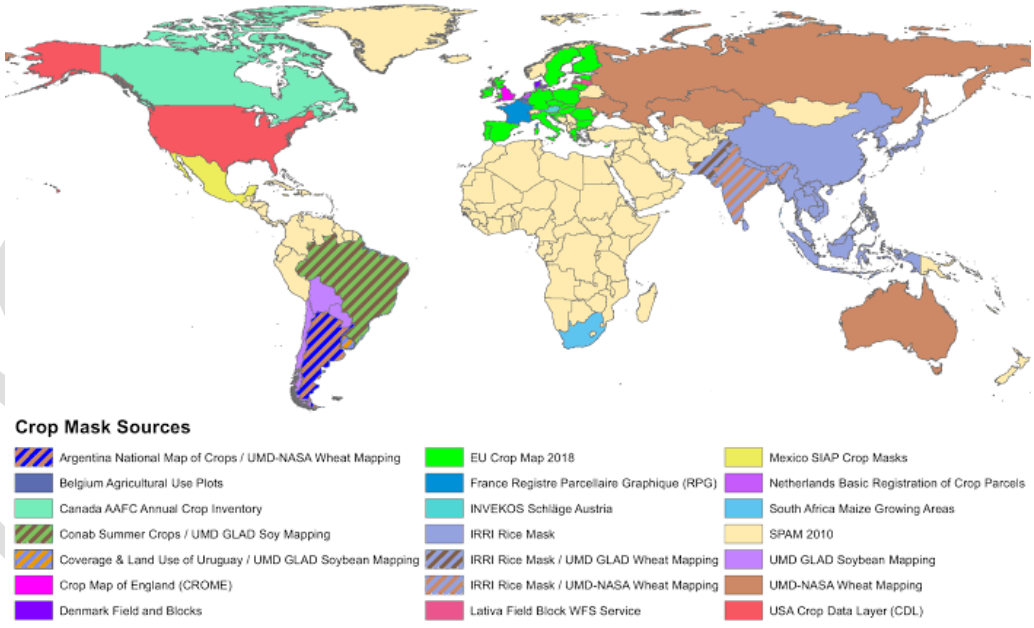


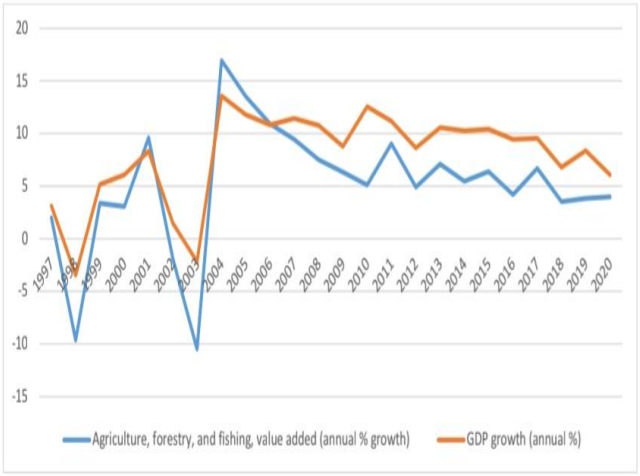
Figure 3 a world map showing major crop producers highlighting Russia's position as a global food producer

ETHIOPIA'S AGRICULTURAL SECTOR

Agriculture is the cornerstone of Ethiopia's economy and society, contributing approximately 35-40% of GDP and employing a staggering 70-80% of the workforce (over 60 million people). It provides the primary food source for the nation's population of over 100 million and generates the lion's share of export earnings, accounting for roughly 70-80% of total exports. Key exports include coffee (generating ~35% of export revenue), oilseeds, pulses, cut flowers, spices, and livestock products.

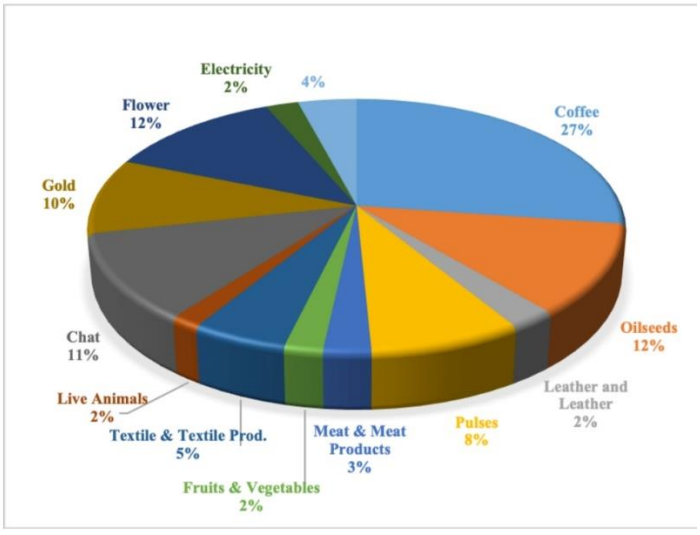
Ethiopia's varied terrain creates distinct farming zones. The sector utilizes an estimated 36-40 million hectares of arable land (though only about 15-16 million hectares are regularly cultivated). The highlands are dominated by smallholder mixed farming, producing staples like teff (covering ~30% of grain area, unique to Ethiopia), wheat, barley, maize, and pulses. Coffee, Ethiopia's iconic export (~3-4% of GDP), thrives here and in mid-altitudes, alongside chat and horticulture. The lowlands (covering ~60% of land area) support pastoralism, home to Africa's largest livestock herd (approx. 70 million cattle, 50 million sheep, 50 million goats, 8 million camels). Enset ("false banana") is a vital perennial food source for ~20 million people in the South.

(Key sources for stats: World Bank, FAO, Ethiopia Central Statistical Agency (CSA), UNDP, Ethiopian government reports - figures represent recent averages/estimates).



Source: Authors Computation using data from NBE, 2020/21

Figure 4 shows Ethiopia's GDP and agriculture growth rates



Source: Authors Computation using MoT and NBE, 2020/21

Figure 5 shows share of major export items

Ethiopia	Russia
Agriculture is a **foundational pillar** of Ethiopia's current development stage.	it is a **strategic sector** within a diversified, industrialized economy.
It is **the largest economic contributor** in terms of GDP Share ranging from~35-40% and it was historically much higher although it is, still dominant). This makes it the largest contributor to economic growth fluctuations in Ethiopia.	It contributes **a little percentage** to the GDP share of ~3-5%. As it is significantly overshadowed by other industries (especially energy) and services. However it acts as a stabilizer and it contributes to stability and diversification away from hydrocarbons.
It employs **the largest work force** of ~70-80% of the total labor force. This makes it a primary source of income and subsistence for the vast majority of the population.	Employment (in the agricultural sector is **low as it employs** ~5-7% of the labor force the sector is highly mechanized and efficient.
Agriculture is the **primary source earner** with major exports being coffee, oilseeds, pulses, flowers, and livestock are major export commodities, generating crucial foreign exchange. However this has made the country's economy highly vulnerable to global commodity price swings and climate impacts on these exports.	In terms of **export earnings Russia is a top global exporter** of wheat, barley, sunflower oil, and fertilizer. This makes agriculture a key source of foreign exchange complementing energy exports. However because of its position as top exporter of various agricultural products Russia is a significant force for geopolitical influence using its food as a tool.
The country has **switched from production** of **food crops** in the recent decades to cash-crop production because of the high economic returns. This has created a challenge in large scale food crops production leaving production basically subsistence. this has been worsened with droughts and climate shocks, leading to frequent food insecurity	Russia's agricultural products **are mainly food** crops and this is very vital for food security which is both achieved and used for strategic goals. This has made Russia self-sustaining and made food security largely achieved for the population. Furthermore, this has been used for advantages of national security and independence. The sector was majorly developed as an import substitution and it proved to be a successful drive

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for large segments of the population due to heavy reliance on rain-fed agriculture. This has left the country dependent on agricultural exports from other countries like Kenya and Russia.	in 2014 after sanctions and ended up boosting self-sufficiency in many staples
Agriculture has been majorly developed as a strategy for reducing widespread rural poverty and a foundation for provision of raw materials (e.g., cotton, leather, oilseeds) for nascent agro-processing industries.	Agriculture was majorly developed as a way of supporting rural development, rural communities and regional development, especially beyond the major cities. It was a strategy aimed at agro-industrial complex development as part of a large, technologically advanced agro-industrial sector (processing, machinery, inputs) and also reduce reliance on energy exports and mitigates economic volatility.
It is however still being constrained by a variety of factors like: • Low productivity (smallholder, rain-fed, limited tech/inputs). • Land degradation and climate vulnerability. • Limited infrastructure (irrigation, storage, transport).	Strengths and focus of the agricultural sector in Russia are: • Vast land resources (especially fertile "Black Earth" region). • Large-scale, mechanized farming (though smallholders exist). • Increasing technological adoption (precision agriculture). • Strong state support and strategic direction ("Food Security Doctrine").

Question 4

Using the above examples, using agriculture as a standard compare the economies of a developed nation and any African nation that is developing. You should take note of the following: **GDP share, Primary role, Engine of Development, Food Security, Export Role Tech Level.**

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AFRICAN COUNTRIES INVESTMENT IN AGRICULTURE

The debate on agricultural investment in Africa involves balancing competing development models, but evidence strongly supports increased strategic investment in agriculture as essential for accelerated growth and human welfare. Here's a breakdown of key arguments:

The Case for MORE Investment

1. **Foundational role in economies**

Agriculture employs ~60% of Africa's workforce (World Bank). Reducing investment risks destabilizing rural livelihoods and increasing urban poverty. It contributes 15-30% of GDP in many countries, and significantly more when including agro-processing. Africa remains a net food importer ($55+ billion annually - FAO). Boosting local production is critical for nutritional security and insulating economies from global price shocks (e.g., Ukraine war impact).

2. High potential for growth and poverty Reduction

African crop yields are often 60-80% below global averages. Targeted investments (irrigation, improved seeds, fertilizer, extension services) offer massive untapped returns. Agricultural growth is 2-4 times more effective in reducing poverty than growth in other sectors (World Bank). It directly raises incomes for the largest group of the poor. Agriculture fuels demand for local services (transport, finance), inputs (fertilizer, tools), and processing industries, creating broader economic multipliers.

3.Catalyst for Industrialization

Agro-processing is often the most viable entry point into manufacturing (textiles, food & beverage, leather). Rising farm incomes generate savings and investment capital for other sectors. A prosperous rural population creates a larger domestic market for manufactured goods and services.

4. Human welfare imperatives

Investing in diverse, sustainable agriculture improves dietary quality and combats malnutrition. Climate-smart investments are vital for adapting to climate change, protecting vulnerable populations.

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Women constitute a majority of smallholder farmers; empowering them through investment (land rights, credit, tech) yields high welfare dividends.

Arguments for less investment

Move people out of farming

Traditional development models prioritize moving labor from low-productivity agriculture to higher-productivity industry/services. However Prematurely abandoning agriculture ignores reality.

Urban job creation cannot absorb Africa's rapidly growing youth population fast enough. Raising agricultural productivity first is necessary to fuel broader growth and make the transition viable/sustainable.

Low returns / Past failures

Historical agricultural projects (e.g., large state farms, inefficient subsidies) had poor returns, discouraging investment.

Past failures highlight the need for better, smarter investment, not less. Focus on smallholders, market linkages, technology, climate resilience, and supportive policies yields high returns. Successes exist (e.g., Ethiopia's agricultural transformation).

3.**Resource curse / focus on minerals**

Resource-rich countries should prioritize extractive industries for faster revenue.

Over-reliance on resources breeds volatility, corruption, and neglects the rural majority. Agriculture provides more stable, inclusive, and job-intensive growth.

How to Invest, Not Just How Much

The debate shouldn't be simply "more vs less," but what KIND of investment:

Shift focus from just feeding families to commercialization, value addition, and market access. Prioritize technology (drought-tolerant seeds, precision farming), infrastructure (storage, roads, irrigation), digital tools (market info, fintech), and research & development.

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Requires sound policies (land tenure security, trade), institutions, finance (credit for smallholders), and climate adaptation strategies.

Target women, youth, and smallholders who form the backbone of African agriculture.

Link farms to agro-industries, input suppliers, and domestic/export markets.

Question 5

With the above points support the view by African union to do massive agriculture development.

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